How a Zipper Holds Shut and Comes Apart

01 What are the main parts of a zipper?



A zipper has three main parts: tape, teeth and slider.

Look closer. Find the two cloth strips, the rows of small teeth on them, and the one slider that rides along the teeth.

A little explanation. A zipper has three main parts. The tape is the cloth strip sewn to the jacket. The teeth, also called elements, are fixed in a row along each tape. The slider is the piece you pull up and down to join or part the teeth.

Try this. Find a zipper on a jacket, bag or pencil case. Point to the tape, the teeth and the slider, and say each name out loud.

02 How do the teeth hold the two sides together?



Each tooth tucks between two teeth on the other side.

Look closer. Compare the open teeth at the bottom with the joined teeth at the top. See how each tooth sits between two teeth on the other side.

A little explanation. Each tooth has a little bump on one side and a hollow on the other. The teeth on the left are set halfway between the teeth on the right. When the slider presses them together, each bump tucks into the hollow of the next tooth, and a joined row of teeth is called the chain.

Try this. Hold your hands flat with fingers spread. Slide them together so each finger fits between two fingers of the other hand. Now pull your hands apart sideways and feel how the fingers hold.

03 Why doesn't a zipper slide open on its own?



With the pull-tab flat, a tiny piece holds the slider still.

Look closer. Look at the pull-tab lying flat against the slider. In the cut-open slider beside it, find the small prong that reaches down toward the teeth.

A little explanation. Many sliders have a lock. When the pull-tab is let go or pressed flat, a tiny piece under the slider slips between the teeth and holds it still. Lifting the pull-tab moves that piece out of the way so the slider can move again.

Try this. Find a zipper with a pull-tab. Let go of the tab and gently tug the two sides of the cloth. Then lift the tab and slide it. Tell what changed.

04 How does a jacket zipper come all the way apart and start again?



The pin slides fully into the box so both rows start lined up.

Look closer. Find the slider waiting on the little box at the bottom of one tape, and the thin pin on the other tape lined up to slide in.

A little explanation. The zipper down the front of a jacket is usually an open-end zipper. One tape ends in a thin insert pin. The other tape ends in a retainer box that holds the slider at the bottom. Pushing the pin fully into the box lines the two rows up, so the slider can start meshing teeth from the very first pair.

Try this. Look at the bottom of a jacket zipper. Find the pin and the box. Watch how the pin goes all the way in before the slider moves up.